

Projective Transport Systems for Finite Power Tables

Divisor-Chain Compatibility and Lift-Fiber Refinement in Finite Residue Rings

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Abstract

A companion paper [9] established fixed-modulus transport closure for the finite power table $T_d(x, k) = x^k \pmod{d}$: rows flow through gcd-depth, columns act as exponent operators, addition requires anchored CRT position, and multiplication folds through saturated gcd-depth before cycling in quotient-unit structure.

This paper treats the cross-modulus problem. Given $d \mid D$, the reduction map

$$\pi_{D,d} : \mathbb{Z}/D\mathbb{Z} \rightarrow \mathbb{Z}/d\mathbb{Z}$$

preserves powers, addition, and multiplication. This is ordinary finite-ring algebra. The purpose of this paper is not to repackage that fact as new ring theory; the purpose is to record its role in the projective system: lower tiles are compressed images of higher tiles, and lift fibers are finite ontology-refinement fibers.

The main structural contribution is the lift-fiber refinement theorem. For $d = \prod p^{\alpha_p}$ and $D = \prod p^{\beta_p}$, each lift fiber decomposes according to the exponent differences $\beta_p - \alpha_p$. If $\alpha_p = 0 < \beta_p$, lifting introduces a new-prime refinement. If $0 < \alpha_p < \beta_p$, lifting deepens an existing prime-power layer. If $\alpha_p = \beta_p$, no new p -information appears. Within the refinement cases, zero-compatible fibers may further split by upstairs valuation depth. Thus projection compresses the tile, while lifting resolves hidden prime-power and CRT-position structure.

The resulting family $\{\mathbb{Z}/d\mathbb{Z}\}$, indexed by divisibility, forms a projective transport system for rows, columns, addition, multiplication, and lift fibers. The paper remains finite-level and ZFC-conservative.

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1 Purpose and scope

Paper [9] was local: it described transport closure inside a single modulus d . This paper is projective: it studies how those finite transport structures behave when one modulus projects to another.

The core question is simple: *If $d \mid D$, does the transport structure on $\mathbb{Z}/D\mathbb{Z}$ project coherently to the transport structure on $\mathbb{Z}/d\mathbb{Z}$?* The answer is yes for the ring operations and power columns:

$$\pi_{D,d}(x + y) = \pi_{D,d}(x) + \pi_{D,d}(y), \quad \pi_{D,d}(xy) = \pi_{D,d}(x) \pi_{D,d}(y), \quad \pi_{D,d}(x^k) = \pi_{D,d}(x)^k.$$

These are standard consequences of modular arithmetic. The specific content here is not that these identities hold; it is how to interpret them: projection compresses a finite transport tile, and lifting resolves the hidden fiber.

The paper has two purposes: (i) to formalize projective compatibility of transport across divisor chains, and (ii) to classify lift fibers as finite ontology refinements. It does not claim novelty in the ring-homomorphism core.

2 Basic setting

For each positive integer d , let $R_d = \mathbb{Z}/d\mathbb{Z}$. The power table at modulus d is the finite object

$$T_d(x, k) = x^k \pmod{d},$$

with row coordinate $x \in R_d$ and column coordinate $k \geq 1$. When $d = 9$, this corresponds to the familiar base-10 digital-root power table, except that the residue 0 (mod 9) is often displayed as 9. In this paper, the algebra is always performed in $\mathbb{Z}/d\mathbb{Z}$.

If $d \mid D$, define the reduction map $\pi_{D,d} : R_D \rightarrow R_d$ by $\pi_{D,d}([x]_D) = [x]_d$. This map is well-defined because congruence modulo D implies congruence modulo d .

3 Coordinate stack

The companion paper used the following finite coordinates. For $x \in R_d$, the gcd-depth is

$$g_d(x) = \gcd(x, d),$$

with $g_d(0) = d$. The surviving quotient modulus is $q_d(x) = d/g_d(x)$. When $q_d(x) > 1$, write $x = g_d(x)u$, where u is a unit modulo $q_d(x)$. The class of u in $(\mathbb{Z}/q_d(x)\mathbb{Z})^\times$ is the quotient-unit coordinate.

The full finite CRT position is the prime-power coordinate vector

$$x \mapsto (x \bmod p^{v_p(d)})_{p \mid d}.$$

It is the finest finite coordinate used in this paper, distinguishing every residue and therefore restoring full transport determinism. Mirror signatures, when used, are reduced coarse summaries of quotient-unit position, especially on odd prime-power components. They are useful but not generally sufficient for additive transport.

4 Lawful divisor chains

A *lawful divisor chain* is a finite chain of positive integers

$$d_0 \mid d_1 \mid \cdots \mid d_n.$$

For such a chain, every projection $\pi_{d_j, d_i} : R_{d_j} \rightarrow R_{d_i}$ is defined whenever $i \leq j$.

A proposed intermediate modulus m is not lawful between D and d unless $d \mid m \mid D$. Thus if $d \nmid m$ or $m \nmid D$, the path $D \rightarrow m \rightarrow d$ is not a projective path in this system. Such a modulus is outside the domain, not a counterexample. This distinction matters because path-independence only makes sense over lawful divisor chains.

5 Relation to the standard inverse system

The family $\{\mathbb{Z}/d\mathbb{Z}\}_{d \geq 1}$ with projection maps $\pi_{D,d}$ for $d \mid D$ is the standard inverse system of residue rings over the divisibility poset. Its inverse limit is the profinite completion $\widehat{\mathbb{Z}}$. This paper does not use the inverse limit and does not make profinite analytic claims. It works only at the finite level, recording how transport and lift-fiber ontology behave under the standard projection maps.

6 Projection preserves rows and columns

Theorem 1 (Power projection compatibility). *Let $d \mid D$. For every $x \in R_D$ and every integer $k \geq 1$,*

$$\pi_{D,d}(x^k) = \pi_{D,d}(x)^k.$$

Equivalently, $\pi_{D,d}(T_D(x, k)) = T_d(\pi_{D,d}(x), k)$.

Proof. Choose an integer representative x . Since reduction modulo d is compatible with multiplication, $[x^k]_d = ([x]_d)^k$. Thus $\pi_{D,d}([x]_D^k) = [x^k]_d = ([x]_d)^k = \pi_{D,d}([x]_D)^k$. So every power column commutes with projection. $\square$

7 Projection preserves addition and multiplication

Theorem 2 (Ring transport projection compatibility). *Let $d \mid D$. For all $x, y \in R_D$,*

$$\pi_{D,d}(x + y) = \pi_{D,d}(x) + \pi_{D,d}(y), \quad \pi_{D,d}(xy) = \pi_{D,d}(x) \pi_{D,d}(y).$$

Proof. Standard quotient-ring homomorphism property. For integer representatives x, y , $[x + y]_d = [x]_d + [y]_d$ and $[xy]_d = [x]_d[y]_d$. $\square$

8 Projective compatibility of transport

Theorem 3 (Transport projection theorem). *Let $d \mid D$. Reduction $\pi_{D,d} : R_D \rightarrow R_d$ commutes with the following transport operations:*

1. row-power transport $x \mapsto x^k$,
2. column action $k \mapsto x^k$,
3. additive transport $(x, y) \mapsto x + y$,
4. multiplicative transport $(x, y) \mapsto xy$.

The lower tile R_d is therefore a transport-compatible projection of the higher tile R_D .

Proof. Power compatibility: Theorem 1. Addition and multiplication: Theorem 2. $\square$

9 Lift fibers

For $d \mid D$ and $r \in R_d$, define the lift fiber

$$F_{D,d}(r) = \{x \in R_D : \pi_{D,d}(x) = r\} = \{x \in R_D : x \equiv r \pmod{d}\}.$$

Theorem 4 (Lift-fiber size). *Let $d \mid D$. Then $|F_{D,d}(r)| = D/d$ for every $r \in R_d$.*

Proof. Write $D = md$. The integer representatives of the fiber are

$$r, r + d, r + 2d, \dots, r + (m - 1)d \pmod{D}.$$

These $m = D/d$ residues are distinct modulo D , and every residue congruent to $r \pmod{d}$ appears in this list. $\square$

Theorem 5 (Lift-fiber partition). *Let $d \mid D$. Then $R_D = \bigsqcup_{r \in R_d} F_{D,d}(r)$.*

Proof. Every element $x \in R_D$ has a unique reduction $r = \pi_{D,d}(x) \in R_d$, so $x \in F_{D,d}(r)$. If $r \neq s$, then no x can satisfy both $x \equiv r \pmod{d}$ and $x \equiv s \pmod{d}$. Hence the fibers are disjoint and exhaustive. $\square$

10 Lift-fiber refinement modes

The previous theorems are standard. The specific contribution begins when we ask what those fibers mean inside the coordinate stack.

Write

$$d = \prod_p p^{\alpha_p}, \quad D = \prod_p p^{\beta_p},$$

where $0 \leq \alpha_p \leq \beta_p$ and only finitely many exponents are nonzero. Each prime component falls into one of three cases:

1. **No change:** $\alpha_p = \beta_p$.
2. **New-prime refinement:** $\alpha_p = 0 < \beta_p$.
3. **Prime-power-depth refinement:** $0 < \alpha_p < \beta_p$.

These are the refinement modes. Zero-compatible fibers may further split by upstairs valuation depth within the second and third cases.

11 Lift-fiber refinement theorem

Theorem 6 (Lift-fiber refinement). *Let $d \mid D$ and write $d = \prod_p p^{\alpha_p}$, $D = \prod_p p^{\beta_p}$. For each $r \in R_d$, the lift fiber $F_{D,d}(r)$ has size D/d and decomposes under CRT into independent prime-power lift coordinates.*

For each prime p , exactly one of the following refinement cases applies:

1. *If $\alpha_p = \beta_p$, then the p -component is already fully visible downstairs and contributes no refinement.*
2. *If $\alpha_p = 0 < \beta_p$, then the lift introduces a new-prime refinement.*
3. *If $0 < \alpha_p < \beta_p$, then the lift deepens an existing prime-power layer.*

Within cases (2) and (3), when the downstairs constraint is compatible with zero at the visible depth, the upstairs lifts may further split according to upstairs valuation depth, producing zero/depth substructure inside the fiber.

Thus every lift fiber is a finite ontology-refinement fiber. Projection compresses prime-power and CRT-position information; lifting restores it in the modes determined by the exponent differences $\beta_p - \alpha_p$.

Proof. By the Chinese Remainder Theorem,

$$R_D \cong \prod_{p \mid D} \mathbb{Z}/p^{\beta_p} \mathbb{Z}, \quad R_d \cong \prod_{p \mid d} \mathbb{Z}/p^{\alpha_p} \mathbb{Z},$$

with the convention that primes not dividing d have $\alpha_p = 0$. The projection $\pi_{D,d}$ acts component-wise:

$$x_p \bmod p^{\beta_p} \mapsto x_p \bmod p^{\alpha_p}.$$

If $\alpha_p = \beta_p$, the upstairs and downstairs p -coordinates agree completely, so no refinement is introduced.

If $\alpha_p = 0 < \beta_p$, the downstairs modulus has no p -coordinate. The lift fiber introduces an entirely new p -component modulo p^{β_p} . This is new-prime refinement.

If $0 < \alpha_p < \beta_p$, the downstairs coordinate records $x_p \bmod p^{\alpha_p}$, while the upstairs coordinate records $x_p \bmod p^{\beta_p}$. The fiber over a fixed downstairs coordinate consists of all compatible higher p -adic residues. This is prime-power-depth refinement.

Within the new-prime and depth-deepening cases, if the downstairs condition fixes divisibility by p to the visible depth, upstairs lifts may separate by higher valuation: a residue that is zero modulo p^{α_p} may lift to residues with valuation exactly α_p , larger valuation, or full zero modulo p^{β_p} . These separate into distinct gcd-depth and quotient-modulus profiles upstairs.

Since the CRT decomposition is independent across primes, the full fiber decomposes as the product of these component refinements. $\square$

12 Example: new-prime refinement, $30 \rightarrow 6$

Let $D = 30 = 2 \cdot 3 \cdot 5$ and $d = 6 = 2 \cdot 3$. Then $\alpha_5 = 0$, $\beta_5 = 1$, so projection $\mathbb{Z}/30\mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}$ forgets the 5-component.

Each residue modulo 6 has exactly $30/6 = 5$ lifts modulo 30. For example, the residue 1 mod 6 lifts to

$$1, 7, 13, 19, 25 \pmod{30}.$$

All are congruent to 1 (mod 6), but they differ in their behavior modulo 5. The downstairs tile cannot see the 5-component; the upstairs tile resolves it. This is the prototype of new-prime refinement.

13 Example: prime-power-depth refinement, $27 \rightarrow 9$

Let $D = 27 = 3^3$ and $d = 9 = 3^2$. Then $\alpha_3 = 2$, $\beta_3 = 3$. No new prime appears; the existing 3-adic coordinate deepens.

Each residue modulo 9 has $27/9 = 3$ lifts modulo 27. For example, 1 mod 9 lifts to

$$1, 10, 19 \pmod{27}.$$

All three agree modulo 9 but differ at the 3^3 level. The lift does not add a new prime component; it increases the resolution of the existing 3-component.

14 Mixed refinement

Some projections contain both modes. For example, $45 \rightarrow 9$ has $45 = 3^2 \cdot 5$, $9 = 3^2$: the 3-depth is unchanged, but the 5-component is introduced; new-prime refinement. By contrast, $90 \rightarrow 30$ has $90 = 2 \cdot 3^2 \cdot 5$, $30 = 2 \cdot 3 \cdot 5$: the prime set is unchanged, but the 3-component deepens from 3^1 to 3^2 ; prime-power-depth refinement. More general projections may combine both effects prime by prime.

15 Path independence

Let $d \mid D \mid E$. There are two ways to project from R_E to R_d : directly via $\pi_{E,d}$, or in two stages via $\pi_{D,d} \circ \pi_{E,D}$. These must agree.

Theorem 7 (Divisor-chain path independence). *If $d \mid D \mid E$, then $\pi_{E,d} = \pi_{D,d} \circ \pi_{E,D}$.*

Proof. For any $x \in R_E$,

$$(\pi_{D,d} \circ \pi_{E,D})(x) = \pi_{D,d}([x]_D) = [x]_d = \pi_{E,d}(x).$$

□

16 Path independence for transport

Theorem 8 (Transport path independence). *Let $d \mid D \mid E$. For every $x, y \in R_E$ and $k \geq 1$, direct projection from E to d agrees with staged projection through D for powers, addition, and multiplication. That is,*

$$\begin{aligned}\pi_{E,d}(x^k) &= \pi_{D,d}(\pi_{E,D}(x)^k), \\ \pi_{E,d}(x + y) &= \pi_{D,d}(\pi_{E,D}(x) + \pi_{E,D}(y)), \\ \pi_{E,d}(xy) &= \pi_{D,d}(\pi_{E,D}(x) \pi_{E,D}(y)).\end{aligned}$$

Proof. By Theorem 7, $\pi_{E,d} = \pi_{D,d} \circ \pi_{E,D}$. By Theorems 1 and 2, each projection preserves powers, addition, and multiplication. Substituting gives the three identities. □

17 Fiber path independence

Theorem 9 (Fiber decomposition across a divisor chain). *Let $d \mid D \mid E$. For $r \in R_d$,*

$$F_{E,d}(r) = \bigsqcup_{s \in F_{D,d}(r)} F_{E,D}(s).$$

Proof. An element $x \in R_E$ lies in $F_{E,d}(r)$ exactly when $\pi_{E,d}(x) = r$. By path independence, $\pi_{E,d}(x) = \pi_{D,d}(\pi_{E,D}(x))$. Let $s = \pi_{E,D}(x)$. Then $s \in F_{D,d}(r)$ and $x \in F_{E,D}(s)$. Equivalently, $F_{E,d}(r)$ is the disjoint union of $F_{E,D}(s)$ as s ranges over $F_{D,d}(r)$, because $\pi_{E,D}$ maps $F_{E,d}(r)$ surjectively onto $F_{D,d}(r)$. Distinct s -fibers are disjoint by definition. □

18 Projective transport system

Theorem 10 (Projective transport system). *The family $\{R_d = \mathbb{Z}/d\mathbb{Z}\}_{d \geq 1}$, indexed by divisibility and equipped with projection maps $\pi_{D,d}$ for $d \mid D$, forms a projective transport system in the following sense:*

1. *Projection preserves the transport operations: row-power transport, exponent-column action, addition, and multiplication.*
2. *Projection has compatible structural fibers: lift fibers have exact size D/d , partition the upstairs tile, and refine the downstairs ontology by prime-power exponent differences.*
3. *Projection is path-independent over lawful divisor chains.*

Proof. Operation preservation: Theorems 1–3. Lift-fiber size and partition: previous theorems. Lift-fiber ontology refinement: refinement theorem. Path independence: Theorem 7 and the transport path-independence and fiber-decomposition theorems. □

19 Audit grid

The theorems above are algebraic. The accompanying certificates serve as reproducibility and regression audits.

Projective transport compatibility. Tested divisor pairs: $(6, 2)$, $(30, 6)$, $(210, 30)$, $(2310, 210)$, $(9, 3)$, $(27, 9)$, $(45, 9)$. Gates: power projection; addition projection; multiplication projection; gcd-depth readout under projection; lift-fiber size. Hard failures: 0.

Lift-fiber refinement. Same divisor pairs; classified fibers as new-prime, quotient-unit, zero/depth, or prime-power-depth. Hard failures: 0.

Projective path independence. Tested divisor chains including $30 \rightarrow 6 \rightarrow 2$, $210 \rightarrow 30 \rightarrow 6$, $2310 \rightarrow 210 \rightarrow 30$, $27 \rightarrow 9 \rightarrow 3$, $45 \rightarrow 9 \rightarrow 3$, $63 \rightarrow 9 \rightarrow 3$. Gates: raw projection; power; addition; multiplication; fiber-size multiplication; fiber partition; gcd-readout. Hard failures: 0.

Multi-path projective coherence. Tested lawful intermediate moduli across multi-path divisor chains. Invalid proposed intermediates were skipped when they failed $d \mid m \mid D$. Hard failures after excluding non-lawful intermediates: 0.

20 Discussion

The mathematical core is standard: $\mathbb{Z}/D\mathbb{Z} \rightarrow \mathbb{Z}/d\mathbb{Z}$ is a ring homomorphism whenever $d \mid D$, so it preserves addition, multiplication, and powers; divisor-chain path independence is function composition.

The value of the paper lies in making explicit several finite-level facts useful for downstream work. A lower tile is not isolated; it is a projection of higher tiles. Projection preserves transport. Lifting is structured, not arbitrary. The lift structure is controlled prime by prime. New-prime refinement and prime-power-depth refinement are different mechanisms. The example $30 \rightarrow 6$ introduces a new 5-component, while $27 \rightarrow 9$ deepens the existing 3-adic layer. That distinction is the structural contribution of this paper.

21 Conclusion

For every divisor relation $d \mid D$, projection $\pi_{D,d} : R_D \rightarrow R_d$ preserves powers, addition, and multiplication. Lift fibers have exact size D/d , partition the upstairs tile, and refine the downstairs ontology according to prime-power exponent differences. For every lawful divisor chain, projection is path-independent.

Projection compresses the tile. Lifting resolves the fiber. Every lawful path down the divisor ladder reaches the same lower tile.

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